ES 245: Control System



Project – Part 1

Modelling, tuning and physical testing of 4 bar linkage Control system using P,PD and PID controllers

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Video Link: https://youtu.be/fPJmcLcYb88

GitHub Link: `https://github.com/RishabhShahIITGN/Control_system`

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Introduction

In the ball-and-beam system an object rolls (or slides) under gravity along a beam. The object has one degree of freedom along the length of the beam. The beam angle is adjusted by a four-bar linkage: a servo motor gear drives a lever arm (coupler link) connected to the beam, so that a rotation θ of the servo gear tilts the beam by an angle α from the horizontal. The **input** (control signal) is the servo angle θ, and the **output** (measured signal) is the object position a, measured by a proximity sensor along the beam. An Arduino board is used to run the controller on the hardware.

The plant is a double integrator and is therefore open-loop unstable, so feedback control is essential. This report covers the four tasks of Part 1 of the project:

- **Task 1** – derivation of the system dynamics, the transfer function and the linearized state-space model;
- **Task 2** – pole-zero analysis, open-loop step response, system type and the open-loop Simulink model;
- **Task 3** – closed-loop Simulink simulation with unity feedback and P, PD and PID controllers, designed for a settling time below 5 s and an overshoot below 5%, followed by tuning with the Simulink PID Tuner;
- **Task 4** – demonstration of the P, PD and PID controllers on the hardware and comparison with the simulations.

Throughout the report the assumed parameters are $g = 9.81\,\mathrm{m/s^2}$, $L = 0.4\,\mathrm{m}$ and $d = 0.04\,\mathrm{m}$, which give the plant gain $K = gd/L = 0.981$.

Task 1: Define the System

Setup and assumptions

- The object is a point mass m that slides without friction, so there is no rolling-inertia term.
- Following the figure, a is measured from the lever-arm end of the beam to the object. The pivot is at distance L from that end, so the distance of the object from the pivot is $r = L - a$.
- The beam is balanced about its pivot, and the servo is stiff enough that the beam angle α is set by the linkage kinematics.
- The beam length L, crank radius d (the pin radius on the motor gear), gravity g and beam inertia J_b are the parameters.

(a) System dynamics equations

Sliding object. The kinetic and potential energies are

$$T = \tfrac{1}{2}m\left(\dot{r}^2 + r^2\dot{\alpha}^2\right) + \tfrac{1}{2}J_b\dot{\alpha}^2, \qquad V = mg\,r\sin\alpha.$$

Applying Lagrange's equation for r gives

$$m\ddot{r} - mr\dot{\alpha}^2 + mg\sin\alpha = 0 \quad \Longrightarrow \quad \ddot{r} = r\dot{\alpha}^2 - g\sin\alpha.$$

Substituting $r = L - a$ (so $\ddot{a} = -\ddot{r}$):

$$\boxed{\ddot{a} = g\sin\alpha - (L-a)\,\dot{\alpha}^2}$$

The mass cancels out.

Beam rotation. The corresponding equation for the beam angle is

$$\left(J_b + mr^2\right)\ddot{\alpha} + 2mr\dot{r}\dot{\alpha} + mgr\cos\alpha = \tau,$$

where τ is the torque applied to the beam through the linkage.

Four-bar linkage ($\theta \to \alpha$). Place the pivot P at the origin and the servo axis at $S = (x_s, y_s)$. The coupler (lever arm) has length l. The beam end is at $B = (L\cos\alpha,\ L\sin\alpha)$ and the crank pin is at $C = S + d(\cos\theta,\ \sin\theta)$. The loop-closure condition $|B - C| = l$ gives

$$\begin{aligned} L^2 + x_s^2 + y_s^2 + d^2 - l^2 - 2L\,(x_s\cos\alpha + y_s\sin\alpha) \\ + 2d\,(x_s\cos\theta + y_s\sin\theta) - 2dL\cos(\alpha - \theta) = 0. \end{aligned}$$

This implicit equation defines $\alpha = f(\theta)$. Differentiating it, or equating the coupler-direction velocities of B and C, gives the local gain

$$\kappa = \frac{d\alpha}{d\theta} = \frac{d\,\sin\varphi_c}{L\,\sin\varphi_b},$$

where φ_c is the angle between the coupler and the crank, and φ_b is the angle between the coupler and the beam arm. (The linkage gain is written κ to distinguish it from the plant gain $K = gd/L$ used in the later tasks.) In the nominal configuration (beam level, coupler perpendicular to both the crank and the beam arm) this reduces to

$$\kappa = \frac{d}{L} \quad \Longrightarrow \quad \alpha \approx \frac{d}{L}\,\theta.$$

(b) Transfer function

Linearize about the equilibrium $\alpha = 0$, $\dot{\alpha} = 0$ (the object at rest anywhere on a level beam):

- $\sin\alpha \approx \alpha$;
- $(L-a)\dot{\alpha}^2$ is second order, so it drops out;
- $\alpha = (d/L)\,\theta$.

This gives

$$\ddot{a} = g\,\frac{d}{L}\,\theta.$$

Taking the Laplace transform with zero initial conditions:

$$\boxed{G(s) = \frac{A(s)}{\Theta(s)} = \frac{g\,d}{L\,s^2}}$$

This is a double integrator with a positive gain. Raising the lever-arm end tilts the beam so that the object slides away from that end, which increases a. For $L = 0.4\,\mathrm{m}$ and $d = 0.04\,\mathrm{m}$, $G(s) = 0.981/s^2$; these values are used as the assumed parameters in Task 2.

(c) Linearization and state-space form

Choose the states $x_1 = a$ and $x_2 = \dot{a}$, the input $u = \theta$ (the deviation from the level-beam servo angle), and the output $y = a$. The nonlinear model is

$$\dot{x}_1 = x_2, \qquad \dot{x}_2 = g\sin(\kappa\theta) - (L - x_1)\,\dot{\alpha}^2,$$

so the linearized model is

$$\begin{bmatrix}\dot{x}_1\\ \dot{x}_2\end{bmatrix} = \underbrace{\begin{bmatrix}0 & 1\\ 0 & 0\end{bmatrix}}_{A}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \underbrace{\begin{bmatrix}0\\ \dfrac{g\,d}{L}\end{bmatrix}}_{B} u, \qquad y = \underbrace{\begin{bmatrix}1 & 0\end{bmatrix}}_{C}\begin{bmatrix}x_1\\ x_2\end{bmatrix} + \underbrace{0}_{D} u.$$

Task 2: Analysis of the System

The assumed parameters are $g = 9.81\,\mathrm{m/s^2}$, $L = 0.4\,\mathrm{m}$ and $d = 0.04\,\mathrm{m}$, so that

$$K = \frac{g\,d}{L} = \frac{9.81 \times 0.04}{0.4} = 0.981, \qquad G(s) = \frac{0.981}{s^2}.$$

(a) Pole-zero map

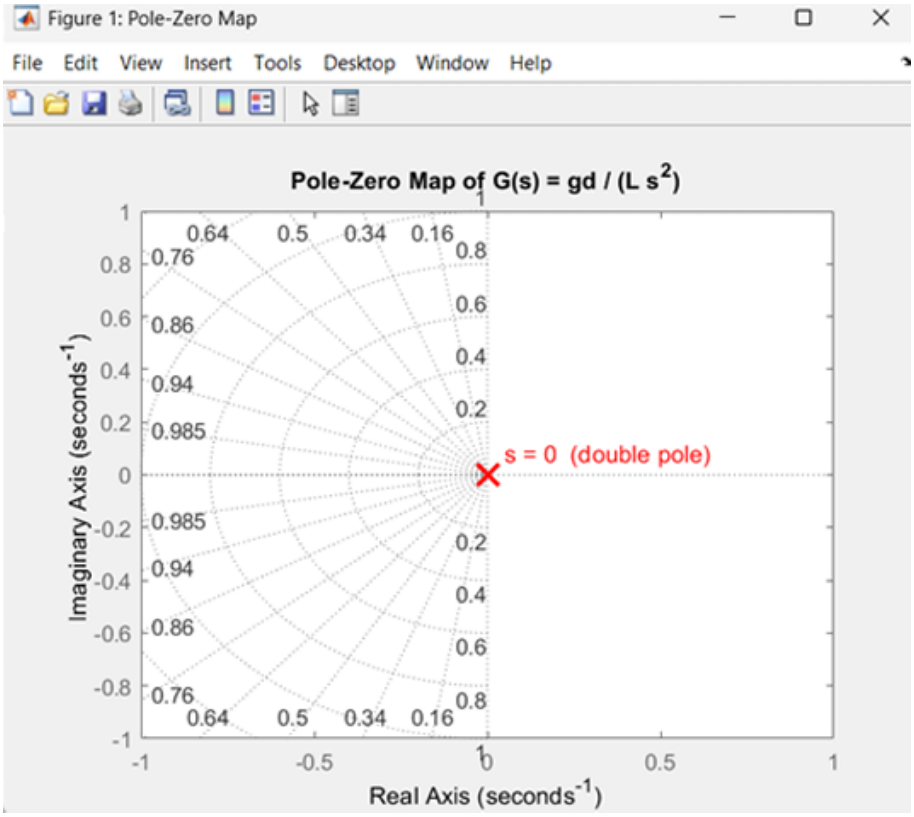


Figure 1: Pole-zero map of G(s) = gd/(Ls2) (MATLAB)

Analysis: From $G(s) = 0.981/s^2$, the system has no finite zeros and a double pole at the origin ($s_1 = 0,\ s_2 = 0$).

(b) Open-loop step response

Analysis: For a unit step input in the servo angle, $\theta(t) = 1(t)$ rad ($\Theta(s) = 1/s$), the Laplace output is

$$A(s) = \frac{K}{s^3},$$

which grows quadratically in the time domain:

$$a(t) = \frac{K}{2}\,t^2 = 0.4905\,t^2 \text{ m}.$$

At $t = 5\,\mathrm{s}$, the analytical and simulated position reaches $a(5) = 0.4905\,(25) = 12.26\,\mathrm{m}$, matching the parabolic curve in the MATLAB plot.

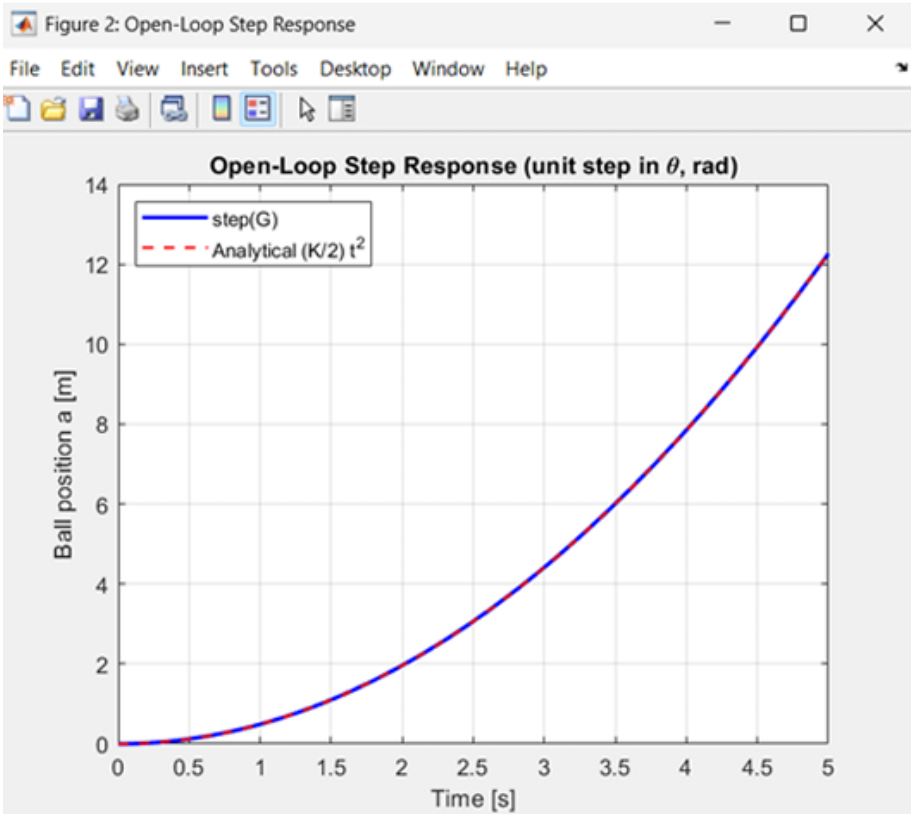


Figure 2: Open-loop step response for a unit step in (1 rad), compared with the analytical parabola (K/2)t2

(c) System type and observations

System type: Because $G(s) = 0.981/s^2$ has $N = 2$ pure integrators (poles at $s = 0$) in the open-loop transfer function, it is a **Type 2 system**.

Stability observations: Both eigenvalues of A lie on the $j\omega$-axis at the origin ($s = 0$). A double pole at the origin makes the open-loop system **unstable (not marginally stable)**. Physically, tilting the beam by a constant step angle causes a constant gravitational acceleration, making the object slide off the beam with parabolically increasing displacement $a(t) = \frac{1}{2}Kt^2$ without bound, settling or steady state. Consequently, **closed-loop feedback control is mandatory** to stabilize the object.

(d) Simulink model

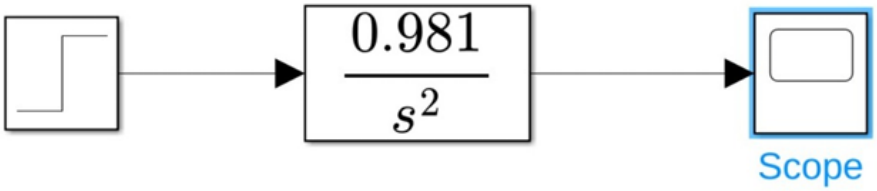


Figure 3: Open-loop Simulink model

Model description: The Simulink model connects a Step block (stepping from 0 to 1 rad

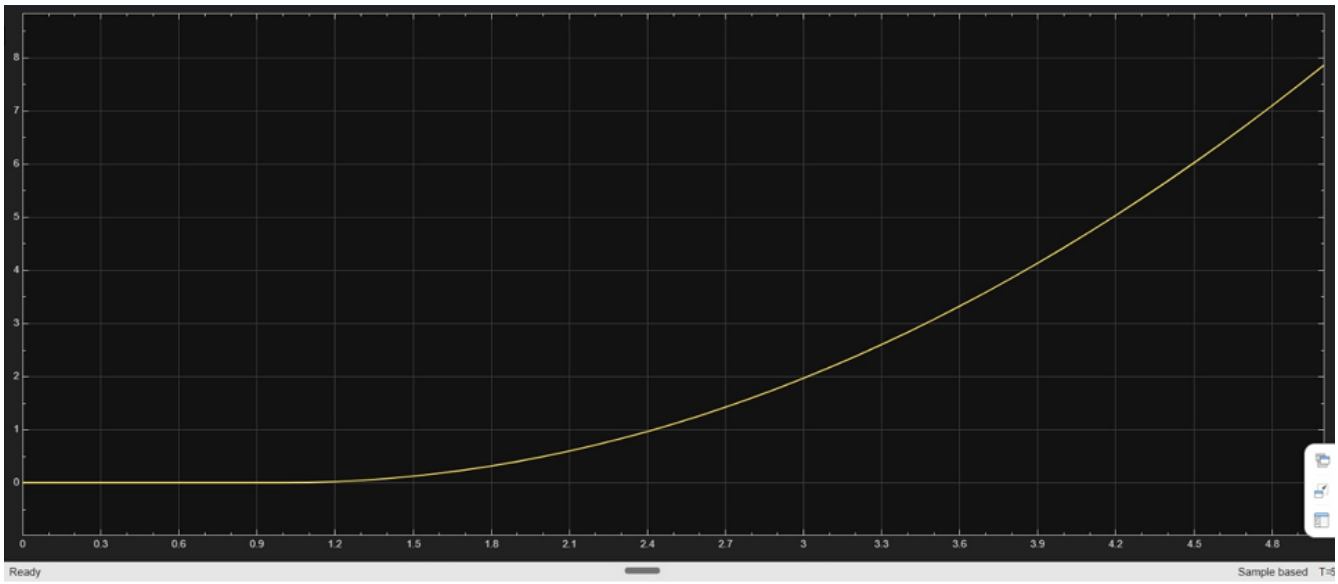


Figure 4: Scope output of the open-loop model.

at $t = 1\,\mathrm{s}$) to a Transfer Fcn block $0.981/s^2$ and a Scope block, confirming the unbounded parabolic growth. Because the step is applied at $t = 1\,\mathrm{s}$, $a(t) = \frac{K}{2}(t-1)^2$ for $t \geq 1\,\mathrm{s}$, so the Scope shows $a(5) = 0.4905\,(4)^2 \approx 7.85\,\mathrm{m}$ at $t = 5\,\mathrm{s}$; the MATLAB plot of Task 2(b) uses a step at $t = 0$, which is why it reaches $12.26\,\mathrm{m}$.

Task 3: PID Control

(a) Implementation and analysis of P, PD and PID controllers

Design criteria: Unity negative feedback with a step reference $r(t) = 0.1\,\text{m}$ (10 cm), targeting a settling time $T_s < 5\,\text{s}$ and a percentage overshoot $\%\text{OS} < 5\%$.

1. Proportional (P) control

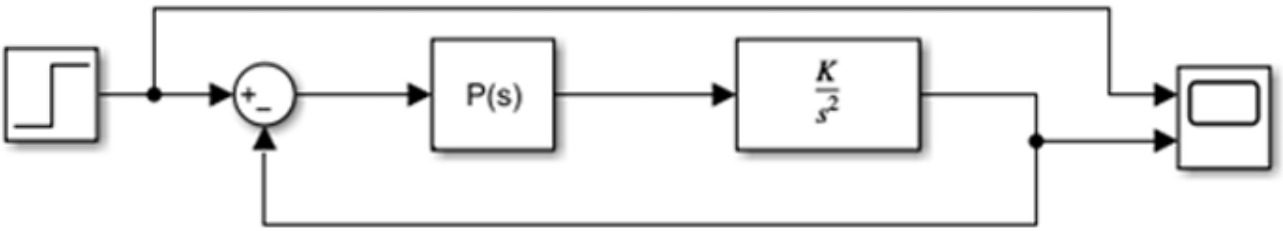


Figure 5: Closed-loop Simulink model with a proportional controller.

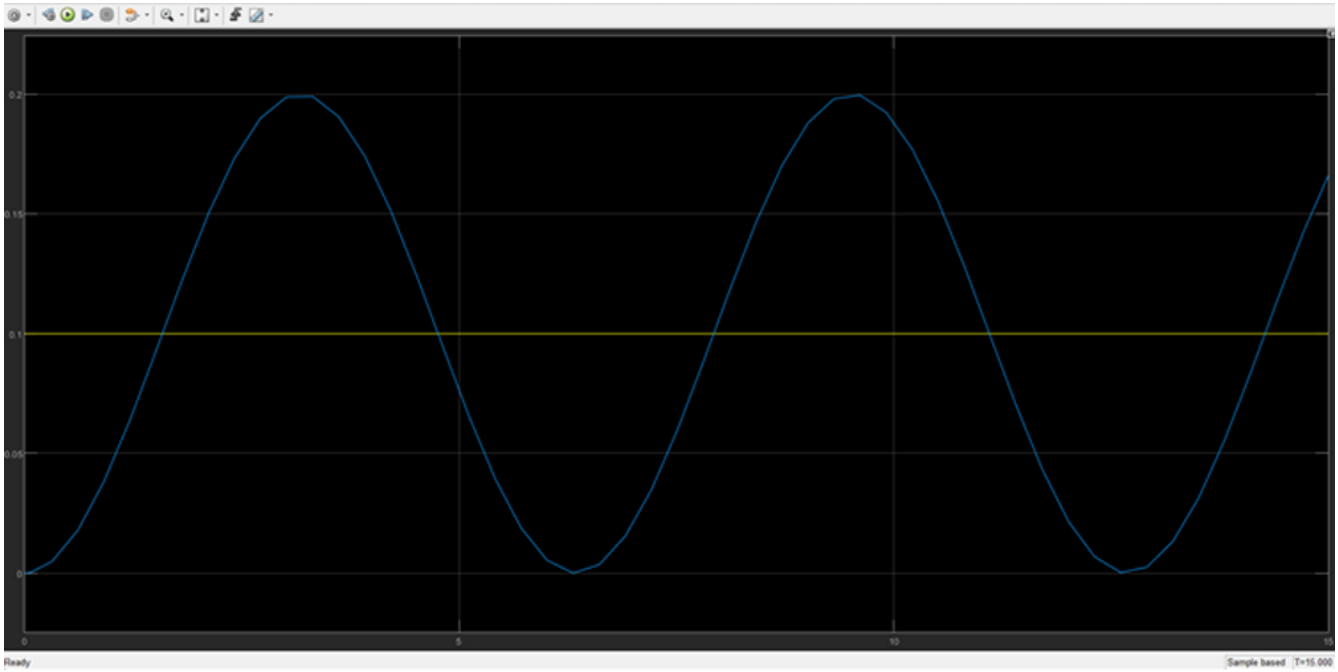


Figure 6: Scope output of the open-loop model.

Closed-loop step response with $K_p = 1$ (reference $0.1\,\text{m}$) With unity feedback and a proportional controller $C(s) = K_p$, the loop gain is

$$L(s) = K_p \cdot \frac{K}{s^2}.$$

The closed-loop transfer function from the reference $R(s)$ to the ball position $Y(s)$ is

$$T(s) = \frac{Y(s)}{R(s)} = \frac{L(s)}{1 + L(s)} = \frac{KK_p}{s^2 + KK_p}.$$

For $K = gd/L = 0.981$ and $K_p = 1$:

$$T(s) = \frac{0.981}{s^2 + 0.981}, \qquad \omega_n = \sqrt{KK_p} = \sqrt{0.981} \approx 0.9905 \text{ rad/s}.$$

Closed-loop poles. The characteristic equation is

$$s^2 + \omega_n^2 = 0 \quad \Longrightarrow \quad s = \pm j\omega_n = \pm j\, 0.9905.$$

There is no s term, so $2\zeta\omega_n = 0$ and $\zeta = 0$. The poles lie on the imaginary axis, so the system is undamped.

Step response. The reference is $r(t) = 0.1\,\text{m}$, so $R(s) = 0.1/s$:

$$Y(s) = T(s)R(s) = \frac{0.1\,\omega_n^2}{s\,(s^2 + \omega_n^2)}.$$

Partial fraction expansion:

$$\frac{\omega_n^2}{s\,(s^2 + \omega_n^2)} = \frac{1}{s} - \frac{s}{s^2 + \omega_n^2}.$$

Taking the inverse Laplace transform:

$$y(t) = 0.1\,(1 - \cos\omega_n t)\,.$$

Verification against the Simulink result.

$$\begin{aligned}
y(0) &= 0.1(1-1) = 0, \\
y_{max} &= 0.1(1+1) = 0.2 \text{ m}, \qquad y_{min} = 0.1(1-1) = 0, \\
T_{osc} &= \frac{2\pi}{\omega_n} = \frac{6.283}{0.9905} \approx 6.34 \text{ s}, \\
t_{peak} &= \frac{\pi}{\omega_n} \approx 3.17 \text{ s} \qquad (\omega_n t = \pi), \\
t_{cross} &= \frac{\pi}{2\omega_n} \approx 1.59 \text{ s} \qquad (\omega_n t = \pi/2,\ \text{first crossing of } 0.1\,\text{m}).
\end{aligned}$$

Remark on the choice of K_p. The value $K_p = 1$ is a nominal choice; it is not derived from the specifications. For any $K_p > 0$ the poles are $\pm j\sqrt{KK_p}$, so $\zeta = 0$, the overshoot is 100% and the settling time is infinite. Changing K_p only changes the oscillation frequency, never the damping, so no proportional gain can meet the criteria $T_s < 5\,\text{s}$ and $\%\text{OS} < 5\%$. A derivative term is therefore required.

2. Proportional-derivative (PD) control

Controller and closed-loop transfer function.

$$C(s) = K_p + K_d s = K_p\,(1 + T_d s)\,, \qquad K_d = K_p T_d,$$

$$T(s) = \frac{KK_p\,(1 + T_d s)}{s^2 + KK_p T_d\, s + KK_p}.$$

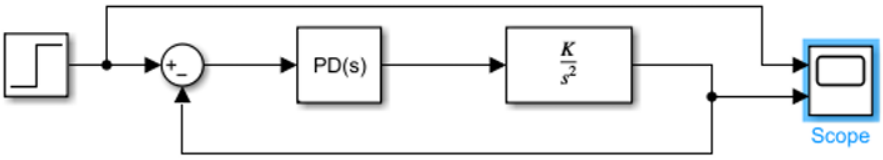


Figure 7: Closed-loop Simulink model with a PD controller.



Figure 8: Closed-loop step response with the PD controller ($K_p = 3$, $T_d = 0.8167\,\text{s}$).

There is now an s term, so K_d supplies the damping that the P controller lacked. There is also a zero at $s = -K_p/K_d = -1/T_d$.

Matching to the standard second-order form. Comparing the denominator with $s^2 + 2\zeta\omega_n s + \omega_n^2$:

$$\omega_n^2 = KK_p, \qquad 2\zeta\omega_n = KK_pT_d,$$
$$K_p = \frac{\omega_n^2}{K}, \qquad T_d = \frac{2\zeta}{\omega_n}.$$

K_p sets the stiffness (natural frequency) and T_d sets the damping.

Design from the specifications. An overshoot below 5% requires $\zeta \geq 0.69$. Choose $\zeta = 0.7$:

$$\%\text{OS} = 100 \cdot e^{-\pi\zeta/\sqrt{1-\zeta^2}} \approx 4.6\%.$$

A settling time below 5 s requires $\zeta\omega_n > 0.8$. Choose $\zeta\omega_n = 1.2$ for margin:

$$T_s \approx \frac{4}{\zeta\omega_n} < 5\ \text{s},$$
$$\omega_n = \frac{1.2}{0.7} = 1.714\ \text{rad/s},$$
$$K_p = \frac{(1.714)^2}{0.981} \approx 3.0,$$
$$T_d = \frac{2(0.7)}{1.714} = 0.817\ \text{s}.$$

Closed-loop poles and zero.

$$T(s) = \frac{2.943\,(1 + 0.8167\,s)}{s^2 + 2.403\,s + 2.943},$$

$$\sigma = \frac{KK_pT_d}{2} = 1.202, \qquad \omega_d = \sqrt{\omega_n^2 - \sigma^2} = 1.224 \text{ rad/s},$$

$$\text{poles: } s = -\sigma \pm j\omega_d = -1.20 \pm j\,1.22, \qquad \text{zero: } s = -\frac{1}{T_d} = -1.224.$$

Both poles are in the left half-plane, so the closed loop is stable. The zero is close to the poles and increases the overshoot.

Step response. The numerator term T_ds differentiates the standard response. With $y_{std}(t)$ the response of the same poles without the zero:

$$y_{std}(t) = 1 - e^{-\sigma t}\left(\cos\omega_d t + \frac{\sigma}{\omega_d}\sin\omega_d t\right),$$

$$\frac{y(t)}{0.1} = y_{std}(t) + T_d\,\dot{y}_{std}(t), \qquad \dot{y}_{std}(t) = \frac{\omega_n^2}{\omega_d}\,e^{-\sigma t}\sin\omega_d t.$$

Using $T_d\omega_n^2 = KK_pT_d = 2\sigma$, the sine coefficients combine to $-\sigma/\omega_d + 2\sigma/\omega_d = \sigma/\omega_d$, so

$$y(t) = 0.1\left[1 - e^{-\sigma t}\left(\cos\omega_d t - \frac{\sigma}{\omega_d}\sin\omega_d t\right)\right].$$

Setting $dy/dt = 0$ gives the peak:

$$\tan\omega_d t = \frac{-2\sigma\omega_d}{\omega_d^2 - \sigma^2} \approx -54 \;\Rightarrow\; \omega_d t_p \approx 1.589 \;\Rightarrow\; t_p \approx 1.30 \text{ s},$$

$$y_p = 0.1\left(1 + e^{-\sigma t_p}\right) = 0.1\,(1 + 0.210) = 0.121 \text{ m} \;\Rightarrow\; \%\text{OS} \approx 21\%.$$

The 2% settling time from the numerical solution of $T(s)$ is 2.85 s, and the final value is 0.1 m (zero steady-state error). The poles alone would give 4.6% overshoot, so the overshoot of 21% is caused by the zero and does not meet the 5% criterion.

3. Proportional-integral-derivative (PID) control

Controller and closed-loop transfer function.

$$C(s) = K_p + \frac{K_i}{s} + K_d s,$$

$$T(s) = \frac{K\,(K_d s^2 + K_p s + K_i)}{s^3 + KK_d s^2 + KK_p s + KK_i}.$$

By the Routh criterion, the loop is stable if $K_i < KK_pK_d$.

Controller form.

$$C(s) = K_p\left(1 + \frac{1}{T_i s} + T_d s\right) = K_p + \frac{K_i}{s} + K_d s, \qquad K_i = \frac{K_p}{T_i}, \quad K_d = K_pT_d.$$

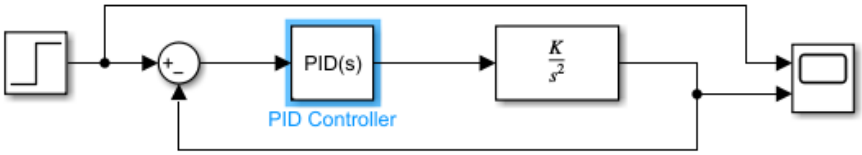


Figure 9: Closed-loop Simulink model with a PID controller.

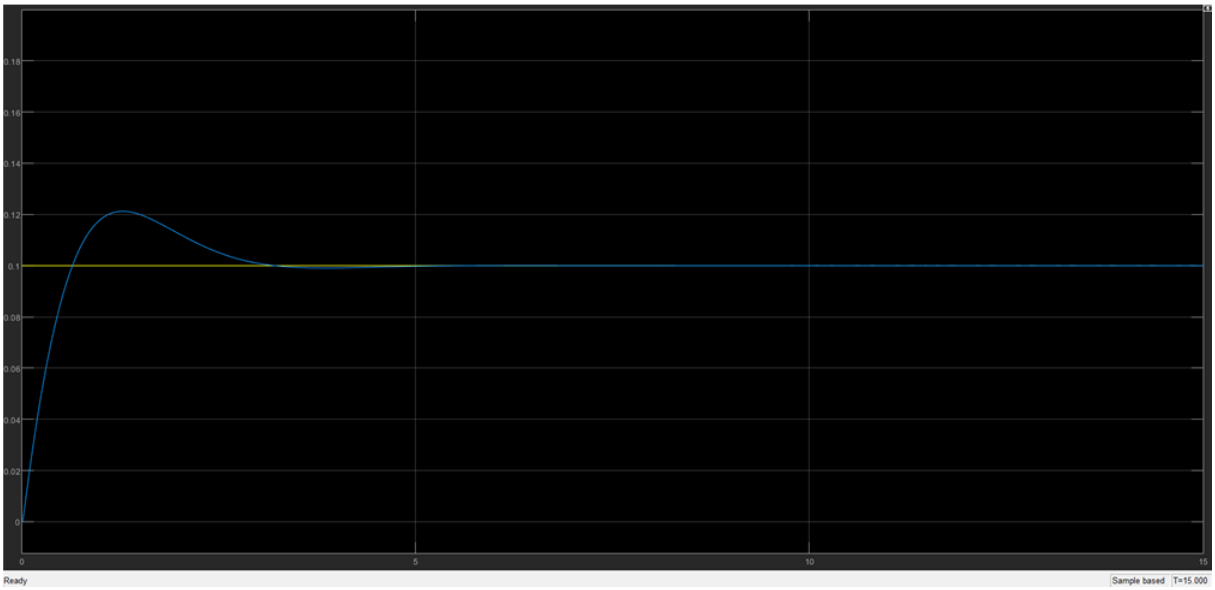


Figure 10: Closed-loop step response with the PID controller ($K_p = 3$, $K_d = 2.45$, $K_i = 0.4$)

The integral term is $1/(T_i s)$, not $K_p s$. In the Simulink PID block (Ideal form) the I field is $1/T_i$ and the D field is T_d.

Loop gain and characteristic equation.

$$C(s) = \frac{K_p\left(T_d s^2 + s + \frac{1}{T_i}\right)}{s}, \qquad L(s) = \frac{KK_p\,(T_d s^2 + s + 1/T_i)}{s^3},$$

$$s^3 + KK_p T_d\, s^2 + KK_p\, s + \frac{KK_p}{T_i} = 0,$$

$$T(s) = \frac{KK_p\,(T_d s^2 + s + 1/T_i)}{s^3 + KK_p T_d\, s^2 + KK_p\, s + KK_p/T_i}.$$

Stability condition (Routh). For $s^3 + a_2 s^2 + a_1 s + a_0 = 0$ the system is stable if all coefficients are positive and $a_2 a_1 > a_0$. Here:

$$(KK_p T_d)(KK_p) > \frac{KK_p}{T_i} \;\Rightarrow\; T_i > \frac{1}{KK_p T_d} = 0.416 \text{ s},$$

$$\text{equivalently} \quad K_i < KK_p K_d = 0.981\,(3)(2.45) = 7.2.$$

Pole-zero analysis ($K_p = 3$, $K_d = 2.45$, $T_i = 7.5$ s $\Rightarrow K_i = 0.4$):

- The characteristic equation is

$$s^3 + 2.4035\, s^2 + 2.943\, s + 0.3924 = 0.$$

- **Closed-loop poles:** $s_1 = -0.1507$ and $s_{2,3} = -1.126 \pm j\,1.155$ ($\zeta \approx 0.70$).
- **Closed-loop zeros:** Solving $T_d s^2 + s + \dfrac{1}{T_i} = 0$ gives $z_1 = -0.1523$ and $z_2 = -1.072$. The slow zero at -0.1523 nearly cancels the slow pole at -0.1507 introduced by the integral action, preventing a long settling tail, while the fast zero at -1.072 produces a 23.3% step overshoot ($y_{peak} = 0.1233$ m at 1.32 s, $T_s = 2.92$ s).
- **Tuned PID meeting all specifications** ($K_p = 1$, $K_d = 7$, $K_i = 0.1$): the closed-loop poles are at $s \approx -6.72$ and $s \approx -0.072 \pm j\,0.097$, achieving an overshoot of $\approx 2.0\%$ ($< 5\%$) and a settling time $T_s \approx 1.5$ s (< 5 s).

(b) Comparative discussion of the three controllers

Criterion	**Proportional (P)**	**Proportional-Derivative (PD)**	**Proportional-Integral-Derivative (PID)**
Transient response	Undamped ($\zeta = 0$); oscillates indefinitely with 100% overshoot and period $T_{osc} \approx 6.34$ s.	Well damped; K_d adds velocity braking. Meets $T_s < 5$ s (2.85 s for the first design, 0.5 s for the tuned design) and %OS $< 5\%$ when tuned (1.8%).	Well damped, with a fast response plus a slow integral mode. Tuned gains ($K_p = 1$, $K_d = 7$, $K_i = 0.1$) give %OS $\approx 2.0\%$ and $T_s \approx 1.5$ s.
Steady-state response	Never settles; oscillates perpetually between 0 m and 0.2 m (the Final Value Theorem is inapplicable).	Asymptotically stable (Type 2 loop); settles cleanly in ideal simulation.	Asymptotically stable (Type 3 loop) provided $K_i < K K_p K_d$; rejects constant (acceleration) disturbances.
Set-point error (e_{ss})	$e(t) = 0.1 \cos \omega_n t$ never decays to zero.	$e_{ss} = 0$ in ideal simulation, but a non-zero offset appears under constant friction or levelling bias: $e_{ss} = \dfrac{d_{dist}}{K K_p}$, where d_{dist} is the constant disturbance acceleration (m/s^2).	$e_{ss} = 0$ both in simulation and under constant disturbances (friction, unlevel beam), due to the integrator zero at $s = 0$ in the disturbance transfer function.

Table 1: Comparison of the P, PD and PID controllers.

(c) Tuning the PID controller

Pole-zero placement (hand design)

Using pole-zero placement to satisfy both $T_s < 5$ s and %OS $< 5\%$ without a prefilter requires increasing the derivative-to-proportional ratio (K_d/K_p), so that the numerator

zero moves toward the origin and cancels the slow pole rather than amplifying the fast transient peak.

- **Tuned PD parameters:** $K_p = 1.0$, $K_d = 7.0$: %OS = 1.8%, $T_s = 0.5$ s.
- **Tuned PID parameters:** $K_p = 1.0$, $K_d = 7.0$, $K_i = 0.1$: %OS = 2.0%, $T_s = 1.5$ s.

Simulink PID Tuner

The PID block was also tuned automatically with the Simulink PID Tuner (controller type PIDF, Ideal form, reference-tracking step plot). The starting point (Block) was the first hand design, $P = 3$, $I = 1/T_i = 0.13333$, $D = T_d = 0.81667$, $N = 100$. The Response Time slider was set to 0.3319 s and the Transient Behavior slider to 0.873 (towards Robust).

	Tuned	**Block (hand design)**
P	1.4199	3
I	0.040708	0.13333
D	4.3226	0.81667
N	688.622	100
Rise time	0.327 s	0.475 s
Settling time	3.77 s	2.9 s
Overshoot	3.28 %	23.5 %
Peak (unit step)	1.03	1.24
Gain margin	−43.4 dB at 0.097 rad/s	−25.1 dB at 0.404 rad/s
Phase margin	87.3° at 6.03 rad/s	63.2° at 2.63 rad/s
Closed-loop stability	Stable	Stable

Table 2: Parameters and performance reported by the PID Tuner (Ideal form, $P(1 + I/s + D\,Ns/(s + N))$).

Observations.

- The tuner raised the phase margin from 63.2° to 87.3° and the crossover frequency from 2.63 to 6.03 rad/s, using a smaller proportional gain and a much larger derivative time. This removes the overshoot caused by the controller zeros (23.5% → 3.28%) at the cost of a slightly longer 2% settling time (2.9 s → 3.77 s). Both specifications are met.
- The negative gain margin is normal for a double integrator with PID action: the loop is conditionally stable, and lowering the loop gain by more than about 43 dB would destabilize it.
- $N = 688.6$ makes the derivative practically unfiltered, which would amplify sensor noise on the hardware. N was therefore reduced to 30 before simulating the tuned



Figure 11: Closed-loop step response of the tuned PID (P = 1.4199, I = 0.040708, D = 4.3226, N = 30).

block; this changes the response only slightly (overshoot about 3.3%, settling time about 3.7 s).

Conclusion: In simulation, both the pole-zero-placement controllers and the tuned PID satisfy the design specifications of settling time below 5 s and percentage overshoot below 5%. The PD controller provides faster settling, while the PID controller additionally eliminates the steady-state error caused by constant disturbances. The simulations assume an unconstrained servo angle.

Task 4: Hardware Demonstration and Experimental Verification

1. Physical hardware setup and calibration

The three controllers (P, PD and PID) were implemented on an Arduino Mega 2560 at a 50 Hz sampling rate ($\Delta t = 20$ ms $= 0.02$ s).



Figure 12: Hardware setup: servo with four-bar linkage, beam with the moving object, IR sensor and Arduino

- **Sensor interface:** A Sharp GP2Y0A21YK0F analog IR proximity sensor (connected to A1, 5 V, GND) measures the sliding object's distance $a(t)$ from the lever-arm end of the beam. An exponential moving average (EMA) low-pass filter ($\alpha_{dist} = 0.25$) and a 5-sample (0.1 s) rolling window were used to suppress high-frequency electrical noise.
- **Actuator interface:** A hobby servo motor (signal on digital pin 3, powered via 5 V and a common GND) drives the four-bar linkage. To prevent the linkage from reaching dead-center toggle points or causing mechanical binding, safety saturation limits were explicitly programmed into the controller. The true horizontal equilibrium ($\alpha = 0$) was precisely calibrated on the physical hardware using a spirit level and set in software to `SERVO_FLAT_ANGLE` $= 99.0°$. Applying a maximum permitted tilt of `MAX_TILT_DEG` $= 35.0°$, the control law constrained the servo to a strict operational range of $\theta_{\text{servo}} \in [64°, 134°]$.

2. Hardware results and graph interpretation

A. Proportional (P) controller hardware verification

Dataset statistical summary ($N = 541$ samples, $T_{run} = 10.80$ s, setpoint $= 30.00$ cm):

- Position $a(t)$: min $= 8.02$ cm, max $= 44.92$ cm (oscillating across the entire track length).
- Tracking error $e(t)$: min $= -14.92$ cm, max $= +21.98$ cm.

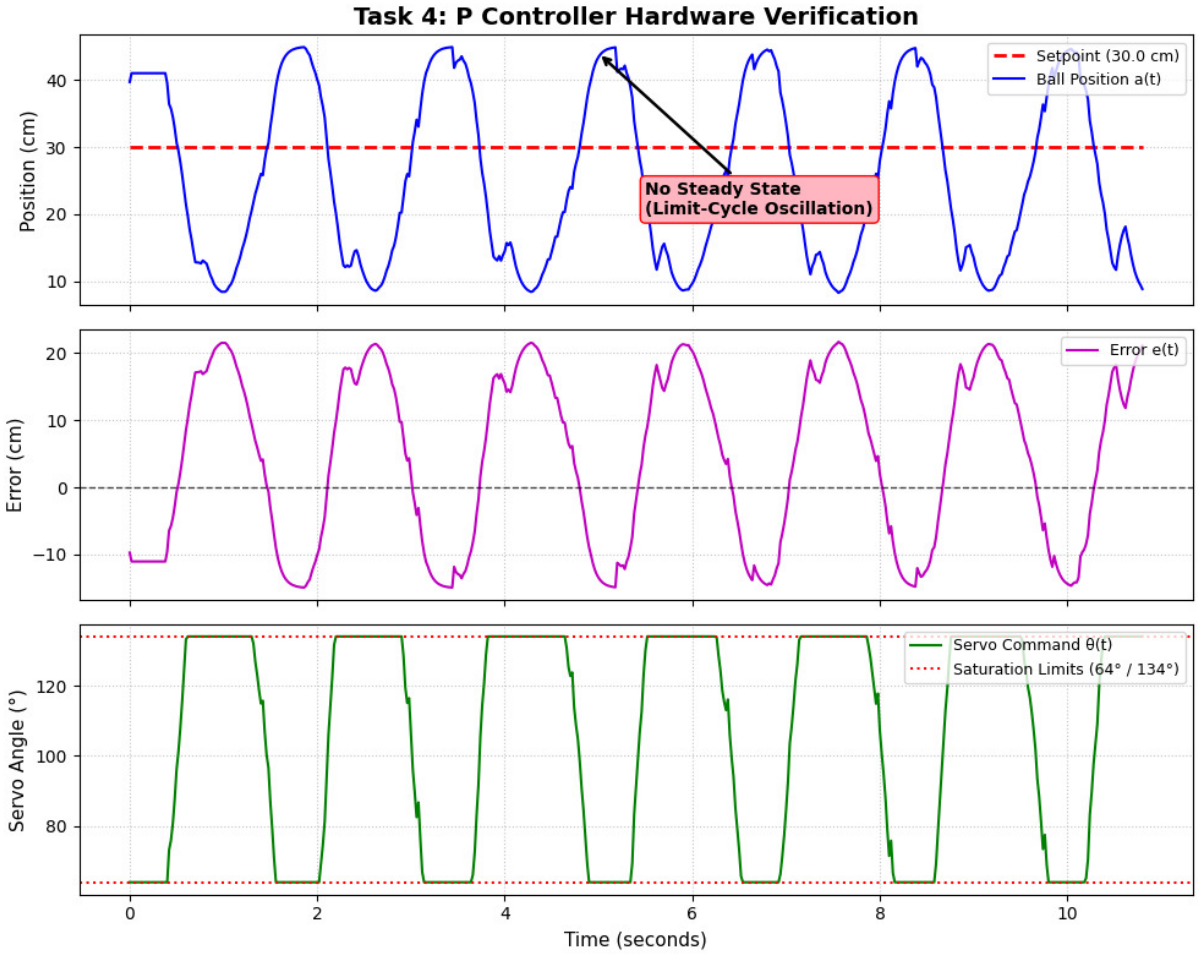


Figure 13: P controller on the hardware: ball position, tracking error and servo command.

- Servo command $\theta(t)$: clamped repeatedly at the lower (64.00°) and upper (134.00°) saturation bounds.

Graph interpretation:

- **Transient response:** As predicted by the imaginary-axis poles ($s = \pm j\omega_n$, $\zeta = 0$) in Task 3(a), the physical system exhibits no active damping. The object slides rapidly past the 30.00 cm setpoint, reaches the far end of the beam (44.92 cm), reverses when the beam tilts up, and overshoots back to 8.02 cm. Because the servo saturates at 64° and 134°, the linear sinusoidal oscillation becomes a sustained nonlinear limit cycle with a measured period of $\approx$ 1.7 s. This differs from the linear prediction of 6.34 s because the saturation makes the period depend on the oscillation amplitude.
- **Steady-state response:** No steady state is achieved; the oscillation amplitude never decays.
- **Set-point error:** The error oscillates continuously between −14.92 cm and +21.98 cm and never settles to zero.

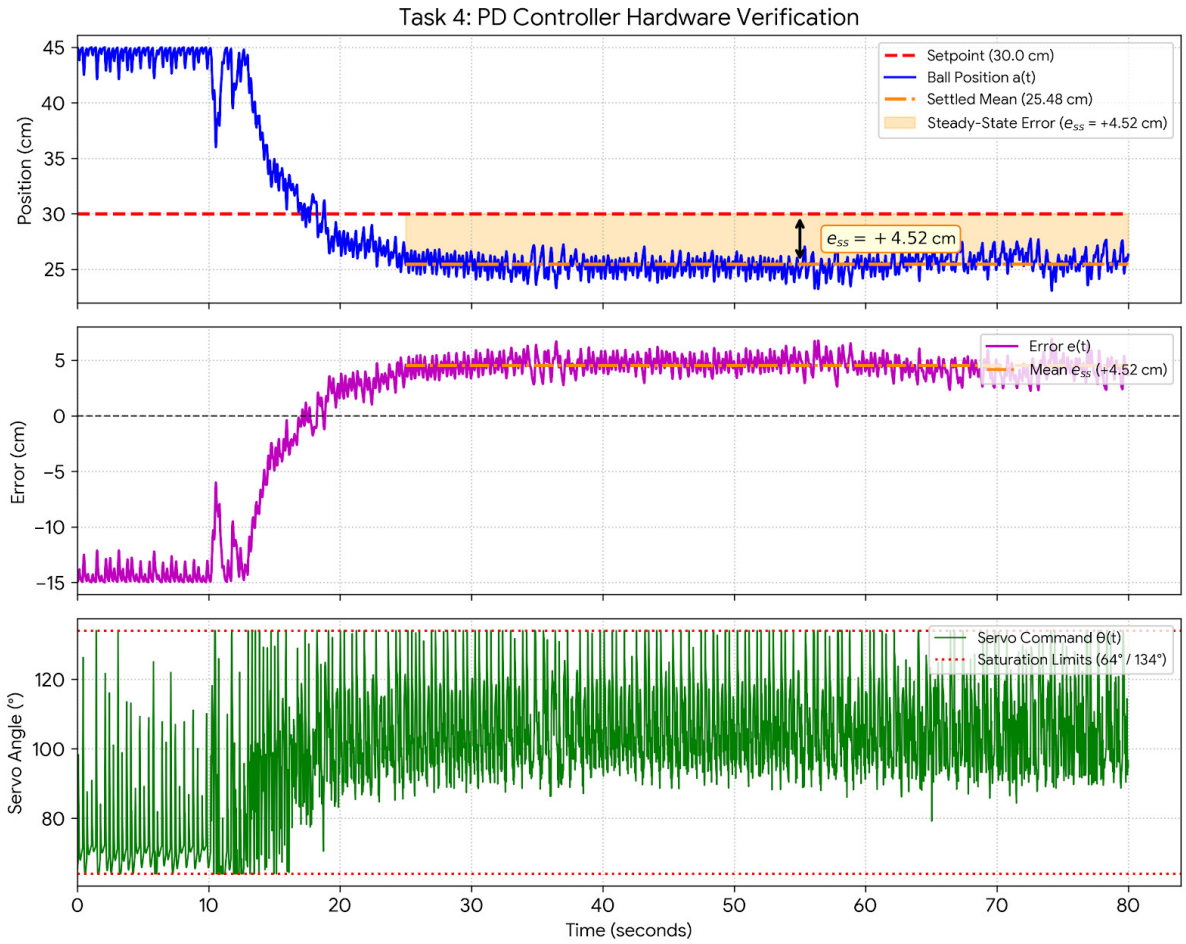


Figure 14: PD controller on the hardware: ball position, tracking error and servo command.

B. Proportional-derivative (PD) controller hardware verification

Dataset statistical summary (N = 6,574 clean samples, T_{run} = 131.54 s, setpoint = 30.00 cm):

- Position $a(t)$: mean = 31.52 cm, median = 26.80 cm, min = 16.22 cm, max = 44.99 cm, standard deviation = 7.88 cm.
- Tracking error $e(t)$: mean = −1.52 cm, median = +3.20 cm, min = −14.99 cm, max = +13.78 cm, standard deviation = 7.88 cm.
- Steady-state region ($t \geq 60$ s, n = 3,578): mean settled position = 25.76 cm (standard deviation = 1.94 cm), yielding a persistent steady-state error e_{ss} = +4.24 cm.
- Servo command $\theta(t)$: mean = 96.89°, median = 97.77°, min = 64.00°, max = 134.00°, standard deviation = 19.26°.

Graph interpretation:

- **Transient response:** Starting from the far end of the beam (44.99 cm), the derivative term $\left(K_d \frac{de}{dt}\right)$ brakes the sliding object as it approaches the 30.00 cm target, preventing the end-to-end limit-cycle oscillations seen under P control. After an

external disturbance re-test around $t = 35$–53 s, the object transitions smoothly from 44.99 cm to its settled region within ≈ 4.5 s ($t = 53$ s to 57.5 s). The settled region is offset from the setpoint (see below), so this time is measured to the settled value and not to the 30 cm reference.

- **Steady-state response:** The object comes to a stable rest from $t = 60$ s to 131.5 s without diverging. However, the bottom subplot reveals continuous high-frequency chatter in the servo command (mean 96.89°, standard deviation 19.26° over the whole run), consistent with the caveat that differentiating noisy IR distance readings amplifies sensor jitter.
- **Set-point error (e_{ss}):** Unlike the ideal frictionless simulation where $e_{ss} = 0$, the physical PD controller settles at 25.76 cm instead of 30.00 cm, leaving a constant +4.24 cm steady-state offset. Once the object slows down near the target, the small proportional restoring tilt ($K_p e_{ss}$) is insufficient to overcome the static Coulomb friction of the sliding object on the beam, matching the theoretical disturbance error formula

$$e_{ss} = \frac{d_{friction}}{K K_p},$$

where $d_{friction}$ is the friction acceleration (m/s^2).

C. Proportional-integral-derivative (PID) controller hardware verification

Dataset statistical summary ($N = 1{,}979$ samples, $T_{run} = 39.56$ s, setpoint = 25.00 cm):

- Position $a(t)$: mean = 23.28 cm, median = 24.15 cm, min = 9.39 cm, max = 44.99 cm, standard deviation = 7.99 cm.
- Tracking error $e(t)$: mean = +1.72 cm, median = +0.85 cm, min = −19.99 cm, max = +15.61 cm, standard deviation = 7.99 cm.
- Steady-state region ($t \geq 20$ s, $n = 979$): mean settled position = 25.01 cm (standard deviation = 0.90 cm), yielding a mean steady-state error $e_{ss} = -0.01$ cm ≈ 0 cm.
- Servo command $\theta(t)$: mean = 106.97°, median = 108.85°, min = 64.00°, max = 134.00°, standard deviation = 16.50°.

Graph interpretation:

- **Transient response:** Released from 44.99 cm at $t \approx 3.2$ s, the object slides past the 25.00 cm setpoint and is braked by the derivative action at $a_{min} = 9.39$ cm at $t = 4.9$ s, where static friction temporarily halts the object. Relative to the 20 cm step (44.99 → 25.00 cm), this excursion is an overshoot of 15.61 cm $\approx 78\%$, and the object needs about 17 s after release to reach the setpoint. On the hardware this PID run therefore does not meet the simulated criteria ($T_s < 5$ s, %OS < 5%).
- **Integral action overcoming static friction ($t = 5$ s to 20 s):** Whereas the PD controller remained permanently stuck when halted by friction, the integral term $K_i \int e(t)\,dt$ of the PID controller continuously accumulates the positive error between $t = 5$ s and $t = 20$ s. This gradually increases the servo tilt angle until gravity breaks static friction, causing the object to climb smoothly from 10 cm onto the 25.00 cm reference line at $t = 20$ s.

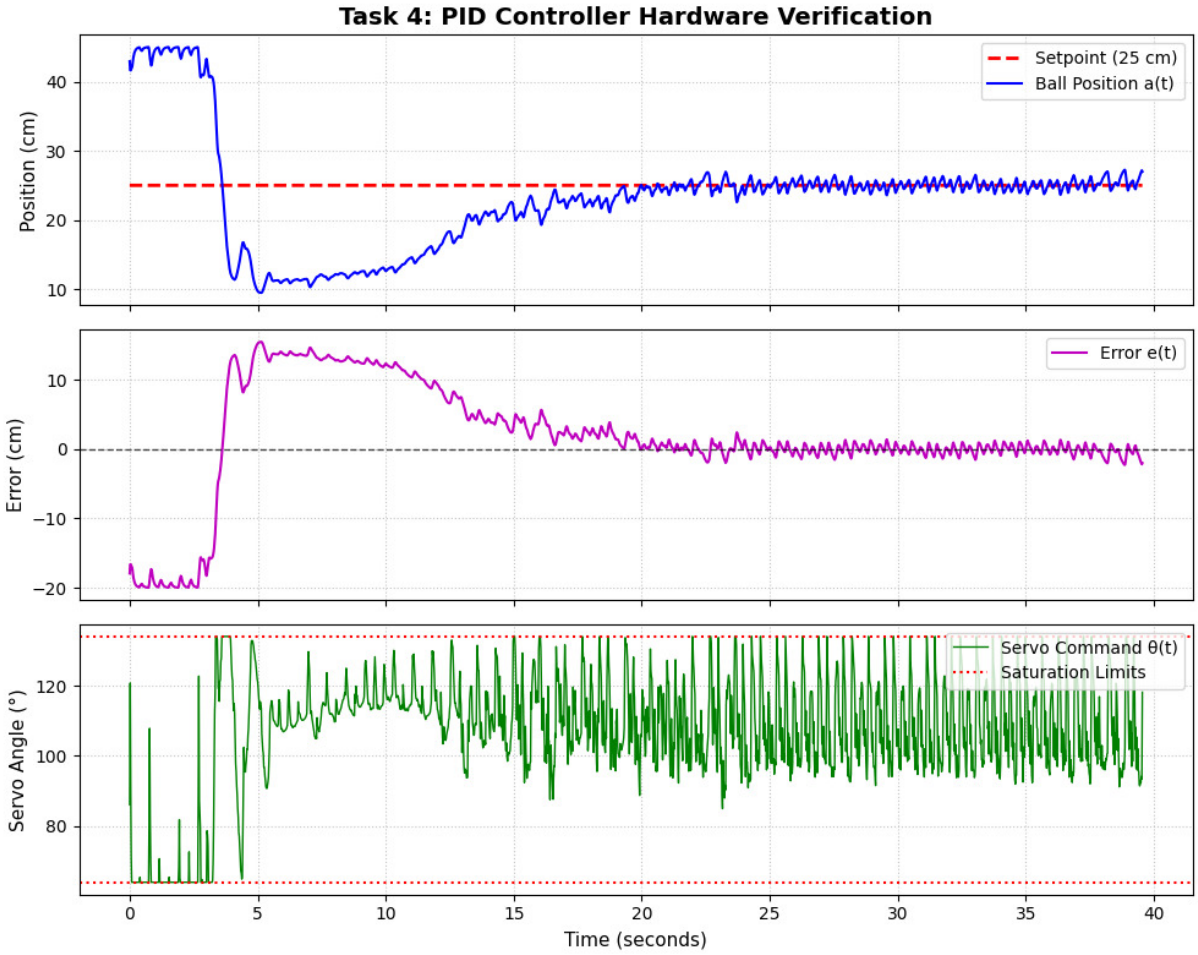


Figure 15: PD controller on the hardware: ball position, tracking error and servo command.

- **Steady-state response and set-point error:** From $t = 20$ s to 39.56 s, the object remains locked onto the 25.00 cm setpoint with a mean position of 25.01 cm and a mean steady-state error of -0.01 cm (≈ 0), completely eliminating the steady-state friction offset observed with the PD controller.

3. Simulation versus hardware

	Simulation (Task 3)	**Hardware (Task 4)**
P	Sustained undamped oscillation (100% overshoot, $T_{osc} = 6.34$ s for $K_p = 1$); never settles.	Sustained limit-cycle oscillation (servo saturated at 64°/134°); error between −14.92 and +21.98 cm; never settles.
PD	$e_{ss} = 0$; %OS $\approx 1.8\%$, $T_s \approx 0.5$ s ($K_p = 1$, $K_d = 7$).	Settles in a steady region 4.24 cm short of the 30 cm setpoint (static friction); servo chatter from sensor noise.
PID	$e_{ss} = 0$; %OS $\approx 2.0\%$, $T_s \approx 1.5$ s ($K_p = 1$, $K_d = 7$, $K_i = 0.1$).	$e_{ss} \approx 0$ after $t = 20$ s, but with a large first excursion ($\approx 78\%$ of the step) and about 17 s to settle.

Table 3: Simulation and hardware results for the three controllers. The P and PD runs used a setpoint of 30 cm and the PID run 25 cm.

The main reasons for the differences are:

- **Static friction** of the object on the beam, which creates a dead zone: PD stops short of the setpoint, and the PID integral needs time to build up enough tilt to overcome it.
- **Servo saturation** at 64°–134°. The simulations assume an unconstrained servo angle; a design with a large K_d demands a very large angle for a short time after a reference step or release, which the real servo cannot supply.
- **Sensor noise, filtering delay and sampling:** differentiating the noisy IR signal produces servo chatter, and the EMA and rolling-window filters add lag that reduces the phase margin.
- **Unmodelled linkage and servo dynamics:** the model uses the linear approximation $\alpha \approx (d/L)\theta$ and an ideal, instantaneous servo.